

2.3 Algorithms

The use of algorithms to describe problems and standard algorithms

2.3.1 Algorithms

- a) Analysis and design of algorithms for a given situation.
- b) The suitability of different algorithms for a given task and data set, in terms of execution time and space.
- c) Comparison of the complexity of algorithms.
- d) Algorithms for the main data structures, (Stacks, queues, trees, linked lists, depth-first (post-order) and breadth-first traversal of trees).
- e) Standard algorithms (Bubble sort, insertion sort, merge sort, quick sort, Dijkstra's shortest path algorithm, A* algorithm, binary search and linear search).